

RADBOUD UNIVERSITY



Faculty of Science

Optimizing Local Movement Rules for Heterogeneous Crowd Evacuation with a Genetic Algorithm

Natural Computing: NWI-IMC042

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1 Problem Statement

Efficient evacuation is an important problem in settings where many people need to move through a limited space under time pressure. In such situations, global crowd behaviour emerges from many local interactions, such as moving toward an exit, avoiding nearby people, and responding to congestion near bottlenecks. Because of this, evacuation is a useful example of collective behaviour in which simple individual rules can lead to complex group-level outcomes such as clogging, uneven flow, and longer evacuation times.

This project focuses on a simplified hospital-inspired evacuation scenario. In hospital-like environments, the crowd is often heterogeneous rather than homogeneous: some individuals can move quickly and independently, while others have reduced mobility and may be more strongly affected by congestion. We model this heterogeneity by distinguishing between *high-mobility* and *low-mobility* agents. This makes the problem scientifically interesting, because a rule set that improves average evacuation performance may not benefit all groups equally. In particular, a strategy that mainly helps fast-moving agents could still perform poorly for slower and more vulnerable individuals.

Existing work has already studied pedestrian evacuation, heterogeneous populations, and optimization of simulation parameters. Evolutionary and related methods have been used, for example, to calibrate pedestrian models or optimize evacuation-related settings. For that reason, our project does not claim to introduce a completely new problem. Instead, we build on this line of work by studying a simple and controlled setting in which a genetic algorithm is used to tune local movement parameters in a heterogeneous crowd model. Our extension is to explicitly evaluate whether such tuned parameters improve evacuation performance in mixed-mobility crowds, and whether they remain effective across different scenarios.

We formulate the evacuation as a multi-objective optimization problem. A real-coded steady-state Genetic Algorithm is used to tune three continuous decision variables:

1. Acceleration factor (how aggressively agents track their preferred speed and heading).
2. Interpersonal repulsion weight (strength of avoidance between agents).
3. Interpersonal repulsion radius (the spatial range of agent-agent interactions).

Candidate solutions are evaluated using multi-seed Monte Carlo averaging to account for stochastic initial placements. The fitness function minimizes a weighted sum of four metrics: total evacuation time, near-collisions, mean congestion at the exit, and a fairness gap (the absolute difference in mean evacuation times between high- and low-mobility groups). Severe penalties are applied if the evacuation is not completed within the time horizon.

Main research question:

Can a real-coded genetic algorithm optimize the interpersonal local movement parameters of a force-based evacuation model to improve global efficiency while minimizing the evacuation-time gap between high- and low-mobility agents?

Sub-questions:

1. Does the GA find continuous parameter sets (acceleration factor, repulsion weight, and radius) that reliably outperform fixed heuristic parameters and random search across multiple stochastic environments?
2. How does the inclusion of a fairness penalty in the fitness function affect the trade-off between global evacuation efficiency (time, congestion, near-collisions) and group-specific delays?

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3. Do the evolved local movement parameters generalize across varying starting layouts, crowd densities, and mobility group ratios without requiring re-optimization?

In summary, this project studies whether evolutionary optimization can improve local movement rules for heterogeneous crowd evacuation in a simple hospital-inspired setting. By focusing on mixed-mobility crowds, group-specific outcomes, and generalization across scenarios, the project aims to provide a clear and feasible contribution that combines collective behaviour modelling with evolutionary algorithms.

2 Related Work

Helbing, D., & Molnár, P. (1995). "Social force model for pedestrian dynamics."[\[4\]](#)

Helbing and Molnár (1995) introduced the social force model, one of the best-known models for simulating pedestrian movement. In their work, pedestrians are influenced by local forces such as attraction toward a goal and repulsion from other pedestrians and walls. This model is important for our project because it shows how simple local interaction rules can lead to complex crowd-level behaviour such as self-organization, congestion, and bottlenecks. We build on this same general idea by using a simplified local-rule crowd model in which agents move toward an exit while interacting with nearby agents. However, rather than only studying the resulting crowd behaviour, we use a genetic algorithm to search for parameter settings that improve evacuation performance in a heterogeneous crowd.

Cui, G., Yanagisawa, D., & Katsuhiro, N. (2022). "A Data Driven Approach to Simulate Pedestrian Competitiveness Using the Social Force Model."[\[2\]](#)

Cui et al. (2022) applied a differential evolutionary algorithm to calibrate the continuous local movements parameters of a social force model to simulate pedestrian dynamics. The algorithm searched for different combinations of parameters, such as desired speed or avoidance strength. The goal was to accurately reproduce the trajectory patterns of "patient" versus "impatient" pedestrians that move through a narrow bottleneck. It relates directly to our project because we will use an EA to optimize local movement parameters in an agent-based pedestrian simulation. We also will encode behavioural differences as parameter sets. However, Cui et al. used the EA to calibrate the algorithm to match the empirical evidence. They did not use it to test whether evolutionary search can be used to improve global evacuation metrics. We will also use an EA to a pedestrian model, but we will use it to search for a set of rules that minimizes evacuation time and congestion.

Cui, G., Yanagisawa, D., & Nishinari, K. (2021). "Incorporating genetic algorithm to optimise initial condition of pedestrian evacuation based on agent aggressiveness."[\[3\]](#)

Cui et al. (2021) incorporated a genetic algorithm into a floor field cellular automata model to study pedestrian evacuation under different levels of agent aggressiveness. Their method searched for initial conditions that improve evacuation efficiency, and they compared the genetic algorithm results with theoretical analysis and handcrafted solutions. This paper is closely related to our project because it also applies a genetic algorithm directly to an evacuation setting rather than only to parameter calibration. However, their work focuses on optimizing initial evacuation conditions and aggressiveness in a floor field model. In our project, we instead focus on optimizing local movement rules in a simplified agent-based model with high- and low-mobility agents. We also explicitly study whether the evolved rules help both mobility groups and whether they generalize across different crowd scenarios.

Alqahtani, A., & Sherif, A. (2025). "Inclusive crowd evacuation modeling under heterogeneous mobility constraints."[\[1\]](#)

Alqahtani and Sherif (2025) applied the social force model to study evacuation dynamics of a heterogeneous crowd that contains normal pedestrians alongside disabled ones (wheel chair and visually impaired agents). They demonstrated that disabled agents significantly increase total evacuation time and congestion. It showed the faster-is-slower effect during a single room evacuation. It relates to our project, we want to simulate a simplified hospital evacuation with a heterogeneous mix of high-mobility and low-mobility agents. In the research of Alqahtani and Sherif they did not explore the underlying interaction rules to mitigate the negative effects. Their parameters are based on empirical evidence. We will also model high- and low-mobility agents, but we add a genetic algorithm to search for the local movement rules that improve overall evacuation, without disadvantaging the slower agents.

3 First experiment analysis:

3.1 Experimental Setup and Parameter Dimensionality

The optimizations space is restricted to three continuous decision variables: acceleration factor, inter-agent repulsion weight, and inter-agent repulsion radius. The wall repulsion strength and interaction radius are fixed at 0.5 and 1.0, respectively.

Candidate parameters are evaluated in a standard room layout containing 30 high-mobility and 10 low-mobility agents, averaged over three stochastic initial-placement replicates (seeds 11, 29, 47). The fitness function minimizes a weighted sum of total evacuation time ($w = 1.0$), near-collisions ($w = 0.05$), mean exit congestion ($w = 1.0$), and the group fairness gap ($w = 2.0$), heavily penalized for incomplete evacuations.

We compare four approaches: a hand-tuned default, two fixed heuristics, a 600-sample uniform random_search, and a steady-state GA (population 24, 25 generations). The GA and random search share an identical computational budget of 600 evaluations.

3.2 Baseline Comparison and Objective Trade-offs

Evolutionary search significantly outperforms the hand-tuned default parameters. The scalar fitness improves from 67.81 (default) to 48.32 (best random) and 46.97 (best GA).

While both search methods drastically reduce the fairness gap (default: 13.4 s $\rightarrow$ random: 7.68 s $\rightarrow$ GA: 7.40 s), analysing the metric breakdown reveals divergent optimization strategies. The best random policy achieves a strictly lower mean total evacuation time (18.1 s) compared to the GA (19.3 s). However, the GA achieves a superior overall fitness by discovering a parameter configuration that yields lower mean congestion (5.27 vs. 5.44), fewer near-collisions (153 vs. 188), and a tighter fairness gap. This demonstrates that the GA does not trivially minimize total time; it effectively exploits the objective to balance global flow with congestion control and group equity.

3.3 Convergence and Evolved Parameters

The GA convergence trajectory demonstrates rapid initial exploitation, dropping from a best fitness of ≈ 55.3 at generation 0 to the low 40s within early generations, followed by diminishing returns. The global optimum of 46.97 is located at generation 23.

The best evolved genome vector is [4.67, 0.61, 2.08]. In the context of the physical model, this corresponds to high acceleration tracking (agents quickly reaching their desired velocity), moderate interpersonal repulsion, and a large interaction radius. This specific parameter combination proactively spaces agents out well before they reach the bottleneck, smoothing the flow and preventing severe localized clustering at the exit.

3.4 Generalization Limits

This experiment validates the optimization pipeline on a static training scenario. Due to the limited sample size (3 seeds) and single room layout, performance variance remains a confounding factor. The current results do not establish whether the evolved parameter vector $[4.67, 0.61, 2.08]$ generalizes. Subsequent experiments require broader Monte Carlo evaluation and testing against unobserved spatial layouts, crowd densities, and altered mobility ratios.

References

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- [2] Geng Cui, Daichi Yanagisawa, and Nishinari Katsuhiro. “A Data Driven Approach to Simulate Pedestrian Competitiveness Using the Social Force Model”. In: *Collective Dynamics* 6 (Feb. 2022), pp. 1–15. DOI: [10.17815/CD.2021.118](https://collective-dynamics.eu/index.php/cod/article/view/A118). URL: <https://collective-dynamics.eu/index.php/cod/article/view/A118>.
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- [4] Dirk Helbing and Péter Molnár. “Social force model for pedestrian dynamics”. In: *Physical Review E* 51.5 (1995), pp. 4282–4286. DOI: [10.1103/PhysRevE.51.4282](https://doi.org/10.1103/PhysRevE.51.4282).